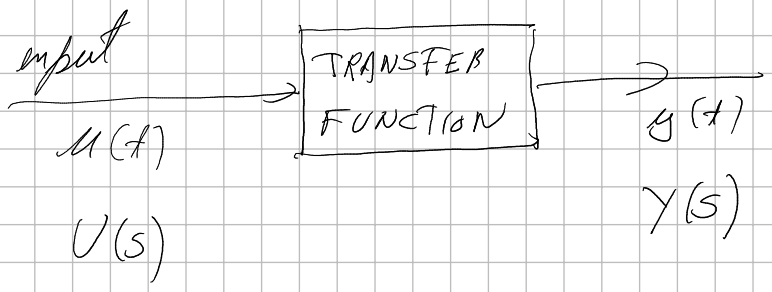
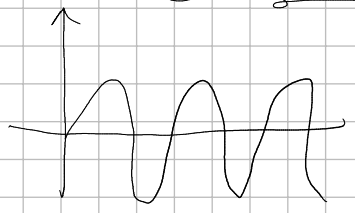
Linear System:



$u(t) = \underline{A \sin(\omega t)} \longrightarrow y(t) = B \sin(\omega t + \phi)$

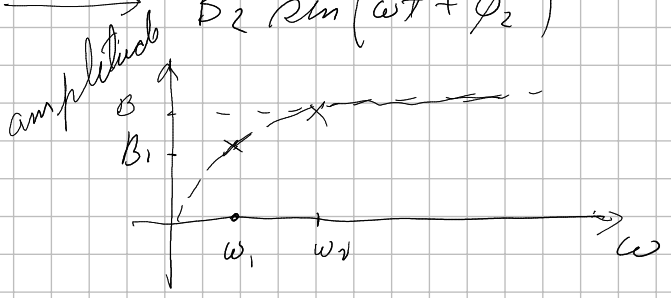
STEADY STATE

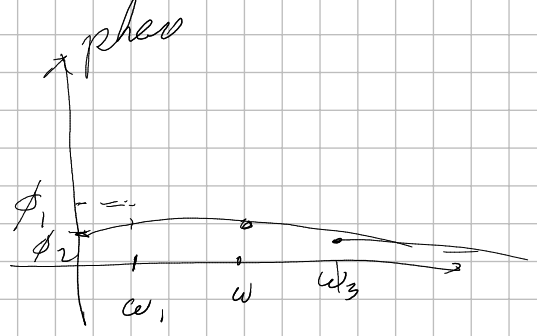




transient steady state

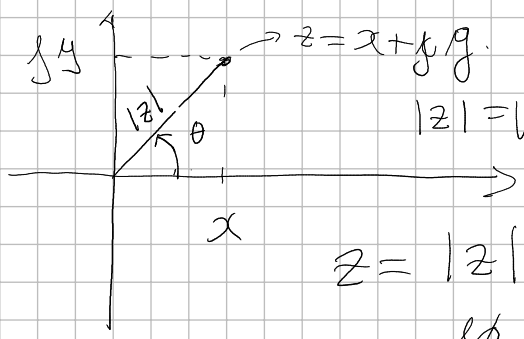
ω	$Y(t)$
ω_1	$\longrightarrow B_1 \sin(\omega t + \phi_1)$
ω_2	$\longrightarrow B_2 \sin(\omega t + \phi_2)$





$G(s) \longrightarrow s = \sigma + j\omega$

$\sigma = 0 \Rightarrow s = j\omega \Rightarrow G(j\omega)$



$$|z| = \sqrt{x^2 + y^2}$$

$$G(j\omega) = Re\left(G(j\omega)\right) + j\,Im\left(G(j\omega)\right)$$

$$z = |z| \angle \theta = |z| e^{j\theta}$$

$$e^{j\phi} = \cos\theta + j\sin\theta$$

$$G(j\omega) = \frac{K}{1 + j\omega T} \qquad |G(j\omega)| = \frac{|K|}{|1 + j\omega T|}$$

$$\left| \underbrace{1}_{Re} + \underbrace{j\omega T}_{Im} \right| = \sqrt{1^2 + (\omega T)^2}$$

$$|K| = K \qquad K > 0$$

$$|G(j\omega)| = \frac{K}{\sqrt{1 + (\omega T)^2}}$$

ω: $\omega_1, \omega_2, \omega_3, \ldots$

$$|G(j\omega_1)| = \frac{K}{\sqrt{1 + (\omega_1 T)^2}}$$

$$\angle G(j\omega) = \angle K - \angle (1 + j\omega T)$$

$$\angle K = 0$$

$$\angle (1 + j\omega T) = \theta = \tan^{-1} \frac{\omega T}{1}$$

$$\angle G(j\omega) = -\tan^{-1} \omega T$$

